

Vortex–Confinement Equations: From Un-normalized Reduced Drift-MHD to the GDB5B Normalization

GDB5B / `vortex_jit` project notes

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Abstract

The `vortex_jit` solver integrates the Beklemishev vortex-confinement model (Beklemishev *et al.*, *Fusion Sci. Technol.* **57** (2010) 351, Eqs. 7–8). Those equations are quoted in the paper *already normalized*, using reference scales specific to the GDT experiment — in particular the potential is normalized to the *applied wall-bias amplitude* ϕ_0 . The GDB5B family of codes uses a different, machine-independent normalization (lengths to L_0 , potential to T_{e0}/e , pressure to $n_0 T_{e0}$). This note (i) re-derives the two evolution equations from the un-normalized reduced drift-MHD system, (ii) applies the *GDB5B* normalization to obtain the dimensionless form, (iii) gives the exact rescaling that maps Beklemishev’s coefficients onto GDB5B ones, and (iv) fixes the physical units of every input in `input.toml`. The central practical result: the *form* of Eqs. 7–8 is invariant, but the numerical values of U , H , κ must be rescaled by the ratio of the potential/pressure reference scales, and user inputs (e.g. ring voltages in volts) are converted to normalized units by dividing by the corresponding base.

1 Notation and geometry

We work in the paraxial, low- β , flute (2-D) approximation of an axisymmetric mirror: perturbations are constant along the field line $\hat{\mathbf{b}}$, and dynamics live in the plane transverse to $\hat{\mathbf{b}}$ with coordinates (x, y) and radius $r^2 = x^2 + y^2$. The uniform guide field is $\mathbf{B} = B_0 \hat{\mathbf{b}}$. The transverse Poisson bracket is

$$\{a, b\} \equiv \partial_x a \partial_y b - \partial_y a \partial_x b, \quad \mathbf{V}_E \cdot \nabla f = \frac{1}{B_0} \{\phi, f\}, \quad (1)$$

where $\mathbf{V}_E = \hat{\mathbf{b}} \times \nabla \phi / B_0$ is the $\mathbf{E} \times \mathbf{B}$ drift velocity (SI units; the paper uses Gaussian units, in which $\mathbf{V}_E = c \hat{\mathbf{b}} \times \nabla \phi / B$ and factors of c decorate the current terms). We evolve two fields: the plasma pressure $p(x, y, t)$ and the electrostatic potential $\phi(x, y, t)$; the generalized vorticity is $\varpi = \nabla_\perp^2 \phi$ (Boussinesq: uniform n , B).

2 Un-normalized reduced drift-MHD model

2.1 Pressure convection (paper Eq. 2)

In the quasi-uniform field the pressure is simply advected by the $\mathbf{E} \times \mathbf{B}$ flow, with small collisional/loss/source corrections:

$$\frac{\partial p}{\partial t} + \mathbf{V}_E \cdot \nabla p = \nu_4^p \nabla_\perp^2 p - \nu_5^p p + q_p, \quad (2)$$

with ν_4^p the transverse (collisional) diffusivity [m²/s], ν_5^p the axial-loss rate [1/s], and q_p the heating source [Pa/s]. Using (1), $\mathbf{V}_E \cdot \nabla p = B_0^{-1} \{\phi, p\}$.

2.2 Charge continuity and the vorticity equation

Quasi-neutrality with axial currents reads $\nabla \cdot \mathbf{j}_\perp + \nabla_\parallel j_\parallel = 0$. Field-line integration over the half-length L of the trap converts the parallel term into a *line-tying* boundary current to the biased end plate (paper Eq. 1),

$$\int \frac{\nabla \cdot \mathbf{j}_\perp}{B} d\ell = -\frac{J_{i0}}{B} \left(1 - \exp\left[\frac{e(\phi_w - \phi)}{T_e}\right]\right) \simeq -\frac{J_{i0}}{B} \frac{e}{T_e} (\phi - \phi_w), \quad (3)$$

where $\phi_w = \phi_{\text{fl}} + \phi_{\text{wall}}$ combines the floating potential and the applied wall bias, and J_{i0} is the equilibrium axial ion current. The transverse current divergence has three physically distinct pieces (paper Eqs. 3–6), obtained from the two-fluid momentum balance following Simakov & Catto:

$$\text{diamagnetic (curvature): } \nabla \cdot \mathbf{j}_d = \nabla \times \frac{\hat{\mathbf{b}}}{B} \cdot \nabla p = \mathcal{K} \{r^2, p\}, \quad (4)$$

$$\text{viscous: } \nabla \cdot \mathbf{j}_\eta = \frac{\eta_i}{enB^2} \nabla_\perp^2 (\nabla \cdot (en \nabla_\perp \phi + \nabla_\perp p_i)), \quad (5)$$

$$\text{polarization (inertia): } \nabla \cdot \mathbf{j}_{pi} \simeq -\frac{1}{B\Omega_i} \left(\partial_t \nabla_\perp^2 (en\phi) + \frac{1}{B} \nabla \cdot \{\phi, en \nabla_\perp \phi\} - \frac{1}{B} \nabla \cdot \{\nabla_\perp \phi, p_i\} + \nabla_\perp^2 s_p \right), \quad (6)$$

with $\mathcal{K} = |\nabla \times (\hat{\mathbf{b}}/B)|/2r$ the field curvature and $\Omega_i = ZeB/m_i$ the ion cyclotron frequency. The polarization piece (6) has been simplified by eliminating $\partial_t \nabla_\perp^2 p_i$ through the pressure equation (2), so that $s_p \equiv \nu_4^p \nabla_\perp^2 p - \nu_5^p p + q_p$ is the right-hand side of the pressure equation itself (paper Eq. 6). Assembling $\nabla \cdot \mathbf{j}_\perp + \nabla_\parallel j_\parallel = 0$, dividing by the polarization prefactor $en/(B\Omega_i) = nm_i/B^2$ (Boussinesq), and writing $\varpi = \nabla_\perp^2 \phi$ gives the *dimensional* vorticity equation

$$\partial_t \varpi + \frac{1}{B_0} \{\phi, \varpi\} - \frac{1}{en} \nabla \cdot \{\nabla_\perp \phi, p_i\} = \frac{B_0^2}{nm_i} \left[\underbrace{-\frac{J_{i0}e}{BT_e L} (\phi - \phi_w)}_{\text{line-tying}} + \underbrace{\mathcal{K} \{r^2, p\}}_{\text{curvature}} + \underbrace{\nabla \cdot \mathbf{j}_\eta}_{\text{viscous}} \right] - \frac{1}{en} \nabla_\perp^2 s_p. \quad (7)$$

The viscous term reduces, under the paper's regularization, to a hyperviscous $-\nu_4 \nabla_\perp^4 \phi$ and a drag $-\nu_5 \nabla_\perp^2 \phi$; the FLR (ion-pressure) polarization term is the $\nabla \cdot \{\nabla_\perp \phi, p_i\}$ contribution on the left, which will carry the coefficient U .

3 GDB5B normalization

3.1 Reference scales

GDB5B normalizes to *machine-independent* plasma scales, not to the applied bias. Let a caret denote a dimensionless quantity, $\phi = \hat{\phi} \phi$ etc.:

Quantity	Base	Value in <code>config.py</code>
length	L_0	L0 (or $\rho_s = c_s/\Omega_i$ if L0 ≤ 0)
potential	$\bar{\phi} = T_{e0}/e$	phi_bar = T0 [V]
pressure	$\bar{P} = n_0 T_{e0}$	P_bar = n0*T0*e [Pa]
field	B_0	B0 [T]
velocity	$\bar{v} = \bar{\phi}/(B_0 L_0)$	$\mathbf{E} \times \mathbf{B}$ reference speed
time	$\bar{t} = B_0 L_0^2 / \bar{\phi} = L_0 / \bar{v}$	t_bar ($\mathbf{E} \times \mathbf{B}$ period)

Here T_{e0} is in energy units and $\bar{\phi} = T_{e0}/e$ is the electron temperature expressed in *volts* (numerically $\bar{\phi}[\text{V}] = T_0[\text{eV}]$).

3.2 Choice of $\bar{t}$ makes advection unity

Insert $\phi = \bar{\phi}\hat{\phi}$, $p = \bar{P}\hat{P}$, $\mathbf{x} = L_0\hat{\mathbf{x}}$, $t = \bar{t}\hat{t}$ into the pressure equation (2). Dividing by $\bar{P}/\bar{t}$,

$$\partial_{\hat{t}}\hat{P} + \frac{\bar{t}\bar{\phi}}{B_0 L_0^2} \{\hat{\phi}, \hat{P}\} = \frac{\bar{t}\nu_4^p}{L_0^2} \hat{\nabla}^2 \hat{P} - \bar{t}\nu_5^p \hat{P} + \frac{\bar{t}}{\bar{P}} q_p. \quad (8)$$

The advection coefficient is $\bar{t}\bar{\phi}/(B_0 L_0^2)$, which equals **1** precisely when $\bar{t} = B_0 L_0^2 / \bar{\phi}$ — the GDB5B time base. The same substitution on the vorticity equation (7), dividing by the polarization prefactor $\frac{n_0 m_i}{B_0^2} \frac{\bar{\phi}}{\bar{t} L_0^2}$, turns the inertia + $\mathbf{E} \times \mathbf{B}$ terms into $\partial_{\hat{t}}\hat{\omega} + \{\hat{\phi}, \hat{\omega}\}$ with unit coefficient (the same cancellation). We recover exactly the paper’s dimensionless form, now in *GDB5B* units:

$$\partial_{\hat{t}}\hat{P} + \{\hat{\phi}, \hat{P}\} = \nu_4^p \hat{\nabla}^2 \hat{P} - \nu_5^p(r) \hat{P} + Q_p, \quad (8)$$

$$\partial_{\hat{t}}\hat{\omega} + \{\hat{\phi}, \hat{\omega}\} - U \hat{\nabla} \cdot \{\hat{\nabla}\hat{\phi}, \hat{P}\} = H(\hat{\phi} - \hat{\phi}_w) + \kappa \{\hat{P}, r^2\} + \nu_4 \hat{\nabla}^2 \hat{\omega} - \nu_5(r) \hat{\omega} - U \hat{\nabla}^2 S_p, \quad (7)$$

with $\hat{\omega} = \hat{\nabla}^2 \hat{\phi}$. **This is the system the code solves.** The dimensionless coefficients are the physical rates/drives multiplied by the appropriate power of the time base:

$$\nu_5^{(p)} = \bar{t} \nu_5^{(p),\text{phys}}, \quad \nu_4^{(p)} = \frac{\bar{t}}{L_0^2} \nu_4^{(p),\text{phys}}, \quad Q_p = \frac{\bar{t}}{\bar{P}} q_p, \quad H \sim \bar{t} \Gamma_{\text{wall}}, \quad \kappa \sim \bar{t}^2 \frac{\mathcal{K}\bar{P}}{n_0 m_i L_0}. \quad (9)$$

The FLR coefficient is especially transparent:

$$U = \frac{p_{i0}}{e n_0 \bar{\phi}} = \frac{p_{i0}}{n_0 T_{e0}} = \frac{T_{i0}}{T_{e0}} \equiv \tau_i \quad (\text{if } p_{i0} = n_0 T_{i0}), \quad (10)$$

an order-unity number in GDB5B units — to be contrasted with Beklemishev’s $U = -5$ to -20 (next section). The detailed machine expressions for H and κ are the paper’s Eqs. (9)–(10) with the substitutions $\tau \rightarrow \bar{t}$ and $\phi_0 \rightarrow \bar{\phi} = T_{e0}/e$.

4 Relation to Beklemishev’s normalization

Beklemishev normalizes the potential to the *applied wall-bias amplitude* ϕ_0 and the pressure to the *on-axis* value p_0 , with time base $\tau = B_0 L_0^2 / \phi_0$ (his lengths use the same flow-layer radius $R_0 \equiv L_0$). Define the two scale ratios

$$\beta_\phi \equiv \frac{\phi_0}{\bar{\phi}} = \frac{e \phi_0}{T_{e0}}, \quad \beta_P \equiv \frac{p_0}{\bar{P}} = \frac{p_0}{n_0 T_{e0}}. \quad (11)$$

The dimensionless fields relate as $\hat{\phi}_G = \beta_\phi \hat{\phi}_B$, $\hat{P}_G = \beta_P \hat{P}_B$, and the time variables as $\hat{t}_G = \hat{t}_B / \beta_\phi$ (since $\bar{t} = \beta_\phi \tau$). Substituting into the Beklemishev-normalized Eqs. (7)–(8) and requiring the advection term to keep unit coefficient (multiply the vorticity equation by β_ϕ^2 , the pressure equation by $\beta_\phi \beta_P$) leaves the *form* unchanged and maps every coefficient:

Coefficient	Beklemishev	× factor	GDB5B
H (line-tying)	H_B	β_ϕ	$H_G = \beta_\phi H_B$
U (FLR)	U_B	β_ϕ / β_P	$U_G = (\beta_\phi / \beta_P) U_B$
κ (curvature)	κ_B	β_ϕ^2 / β_P	$\kappa_G = (\beta_\phi^2 / \beta_P) \kappa_B$
ν_4, ν_5	–	β_ϕ	$\nu_G = \beta_\phi \nu_B$
ν_4^p, ν_5^p	–	β_ϕ	$\nu_G^p = \beta_\phi \nu_B^p$
Q_p	–	$\beta_\phi \beta_P$	$Q_{p,G} = \beta_\phi \beta_P Q_{p,B}$
$\hat{\phi}_w$ (wall bias)	$O(1)$	β_ϕ	$\hat{\phi}_{w,G} = \beta_\phi \hat{\phi}_{w,B} = e V_{\text{ring}} / T_{e0}$

Consistency condition. If the machine is such that $\phi_0 = T_{e0}/e$ ($\beta_\phi = 1$) and $p_0 = n_0 T_{e0}$ ($\beta_P = 1$), every factor is unity and **Beklemishev’s tabulated values carry over unchanged**: $H = 10$, $\kappa = 1$, $U = 0$ (Fig. 5) or $U = -5$ (Fig. 6), $\nu_5 = 0.02$, $\nu_4 = 0.002$. This is the default assumption baked into `input.toml`. When your machine’s applied bias ϕ_0 differs from T_{e0}/e , multiply as in the table. Concretely, Beklemishev’s large $|U| = 5\text{--}20$ is *not* a large physical FLR effect — it is $U_G \sim \tau_i \sim 1$ divided by $\beta_\phi = e\phi_0/T_{e0} \ll 1$; his bias amplitude ϕ_0 is a small fraction of T_{e0}/e , which inflates the bookkeeping value of U .

5 GDT vs. WHAM: physical coefficient values

The dimensionless coefficients U, H, κ, ν are *machine-specific*. Beklemishev’s tabulated values ($U = -5$, $H = 10$, $\kappa = 1$, $\nu_5 = 0.02$, $\nu_4 = 0.002$) reproduce his GDT figures, but a different device requires recomputing them from its parameters through the GDB5B-normalized definitions of Secs. 3–4. We estimate them here for WHAM (the Wisconsin HTS Axisymmetric Mirror; Endrizzi *et al.* 2023; Frank *et al.* 2026).

WHAM reference bases. With $B_0 = 0.86\text{ T}$, $n_0 = 2 \times 10^{19}\text{ m}^{-3}$, $T_{e0} = 1\text{ keV}$, $L_0 = 0.2\text{ m}$, deuterium ($A_i = 2$):

$$\bar{t} = \frac{B_0 L_0^2}{\bar{\phi}} = 34.4\text{ }\mu\text{s}, \quad \bar{\phi} = \frac{T_{e0}}{e} = 1000\text{ V}, \quad \bar{P} = n_0 T_{e0} = 3.2\text{ kPa}, \quad \rho_s = 5.3\text{ mm}.$$

Coefficient	WHAM estimate	Physical basis	vs. GDT
U (FLR)	≈ -10 (-5 to -20)	$U = T_{i0}/T_{e0}$; fast ions ~ 10 keV over $T_e \sim 1$ keV	larger
$\hat{\phi}_w$ (bias)	up to ≈ 2.5	2.5 kV rings / $\bar{\phi} = 1$ kV	$\sim 2.5\times$ deeper
ν_5, ν_5^p	≈ 0.03 – 0.7	$\bar{t}/\tau_{\parallel}$; Pastukhov ($\tau \sim \text{ms}$) to gas-dynamic ($\tau \sim 50 \mu\text{s}$)	uncertain
κ (curvature)	≈ 1 (sign +)	mirror bad curvature (on-axis hill)	similar
H (line-tying)	$\approx \mathcal{O}(10)$	biased-limiter sheath current $\times \bar{t}$	similar
ν_4, ν_4^p	≈ 0.001 – 0.002	regularization ($D_{\perp} \bar{t}/L_0^2$; classical smaller)	same

What differs from the paper defaults.

1. $U \approx -10$, **not** -5 . WHAM’s fast/sloshing ions (~ 10 keV, from 25 keV NBI) over $T_e \sim 1$ keV give $T_i/T_e \sim 10$, so FLR is *stronger* than the Fig. 5/6 default (larger still with the 25 keV injection energy).
2. **Bias** $\hat{\phi}_w$ **up to** ~ 2.5 , not the $\mathcal{O}(1)$ “unit bowl.” WHAM’s 2.5 kV rings over $\bar{\phi} = 1$ kV drive a $\sim 2.5\times$ deeper potential well, hence stronger sheared $E \times B$ rotation — the knob that most directly sets the vortex.
3. ν_5 **spans an order of magnitude**. Since $\nu_5 = \bar{t}/\tau_{\parallel}$, it depends on the confinement regime: Pastukhov/ambipolar ($\tau_{\parallel} \sim \text{ms} \Rightarrow \nu_5 \sim 0.03$, near the paper value) vs. gas-dynamic ($\tau_{\parallel} \sim 50 \mu\text{s} \Rightarrow \nu_5 \sim 0.7$, axial loss *faster* than the $E \times B$ time). Behind the limiter it is $\times 20$ regardless.

Caveats. κ and H are order-of-magnitude only: κ needs WHAM’s $1/B(r, z)$ curvature profile (and its *sign*, here assumed unfavorable), H the end-ring sheath-current model. U and ν_5 carry genuine physics uncertainty (non-thermal T_i ; confinement regime), not just normalization book-keeping. The order-unity defaults are a sound *qualitative* starting point; a *quantitative* WHAM run should use $U \approx -10$, $\hat{\phi}_w$ up to 2.5, and ν_5 chosen for the intended confinement regime.

6 Units of the input file

The solver runs entirely in the GDB5B-normalized variables of Eqs. (8)–(7). Physical inputs are converted at load time using the bases of Sec. 3. The [normalization] section sets B_0, n_0, T_{e0}, L_0 and hence every base.

Input (<code>input.toml</code>)	Physical unit	Normalized by	Note
<code>[bias] ring_volts</code>	volts [V]	$\bar{\phi} = T_0$	$\hat{\phi}_w = V_{\text{ring}}/T_0$
<code>[bias] ring_radII</code>	L_0 (or m)	L_0 if physical	ring outer radii
<code>[domain] Lx, Ly</code>	L_0 (or m)	L_0 if physical	box size
<code>limiter_radius</code>	L_0 (or m)	L_0 if physical	limiter at $r = 1.5 L_0$ (paper)
<code>[initial] pres_amp</code>	$\bar{P}$ (or Pa)	$\bar{P}$ if physical	on-axis pressure
<code>pres_width</code>	L_0 (or m)	L_0 if physical	profile width
<code>U,H,kappa</code>	dimensionless	—	see Sec. 4 table
<code>nu4,nu5,nu4p,nu5p</code>	dimensionless	$(\bar{t}, \bar{t}/L_0^2)$	or give physical rate $\times \bar{t}$
<code>[time] dt</code>	$\bar{t}$	$\bar{t}$	physical $\Delta t = dt \cdot \bar{t}$

The code exposes explicit unit switches so the user can enter machine values directly: `[bias] units = "volts"` (default; divide by $\bar{\phi}$) or `"normalized"`; `[domain] length_units = "L0"` (default) or `"m"` (divide lengths by L_0); `[initial] pres_units = "normalized"` (default) or `"Pa"` (divide by $\bar{P}$).

7 Cross-check against the implementation

Each RHS term in `stepper.py` maps one-to-one onto Eqs. (8)–(7):

Term	Equation	<code>stepper.py</code>
$-\{\hat{\phi}, \hat{P}\}$	(8)	<code>-akw(phi,pres,kp)</code>
$+\nu_4^p \hat{\nabla}^2 \hat{P}$	(8)	<code>+nu4p*delsq(pres,kp)</code>
$-\nu_5^p(r) \hat{P}$	(8)	<code>-nu5p_field*pres</code>
$+Q_p$	(8)	<code>+Qp</code>
$-\{\hat{\phi}, \hat{\omega}\}$	(7)	<code>-akw(phi,vort,kp)</code>
$+H(\hat{\phi} - \hat{\phi}_w)$	(7)	<code>+H*(phi-phi_w)</code>
$+\kappa\{\hat{P}, r^2\}$	(7)	<code>+kappa*akw(pres,r2,kp)</code>
$+\nu_4 \hat{\nabla}^2 \hat{\omega}$	(7)	<code>+nu4*delsq(vort,kp)</code>
$-\nu_5(r) \hat{\omega}$	(7)	<code>-nu5_field*vort</code>
$+U \hat{\nabla} \cdot \{\hat{\nabla} \hat{\phi}, \hat{P}\}$	(7)	<code>U*(derx(akw(derx(phi),pres))+dery(akw(dery(phi),pres)))</code>
$-U \hat{\nabla}^2 S_p$	(7)	<code>U_source = -U*delsq(Qp)</code>

The FLR term uses the identity $\nabla \cdot \{\nabla \phi, P\} = \partial_x \{\phi_x, P\} + \partial_y \{\phi_y, P\}$ (no third derivatives), and $S_p \approx Q_p(r)$ as in the paper. The recovery $\hat{\omega} = \hat{\nabla}^2 \hat{\phi} \Rightarrow \hat{\phi}$ is the periodic FFT Poisson solve `getphi`. Grid-scale regularization is provided solely by the *physical* ν_4, ν_4^p terms; there is no additional numerical hyperdiffusion.

Implementation note (unit conversion). The wall-bias potential $\hat{\phi}_w$ enters through the line-tying term $H(\hat{\phi} - \hat{\phi}_w)$, so ring voltages *must* be in the same (normalized) units as $\hat{\phi}$. With `[bias] units = "volts"` the loader divides `ring_volts` by $\bar{\phi} = T_0$ before building `phi_w`; lengths and pressure inputs are converted analogously when their `*_units` switch is set to physical. This makes the equilibrium bowl depth $\hat{\phi}_w \sim O(1)$ correspond to an applied bias $\sim T_{e0}/e$, consistent with Sec. 4.